

ORF307: Optimization

Precept: #2

Agenda

- Ballistic example
- Inverse of a matrix with LU factorization
- More plotting examples (see notebooks in Companion)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Ballistic example

Overview:

- a projectile is moving in a 2-dim space
- we sample its position and velocity at times $\tau = 0, 1h, 2h, \dots, Th$ (so we sample $T + 1$ times)
- $p_t \in \mathbb{R}^2$ is the position at time $\tau = th$
- $v_t \in \mathbb{R}^2$ is the velocity at time $\tau = th$
- $f_t \in \mathbb{R}^2$ is the total force at time $\tau = th$
- $x_t = (p_t, v_t)^T \in \mathbb{R}^4$ is the projectile's state at time $\tau = th$

Formal model:

- $f_t = mg - \eta v_t$ (this is a common approximation for a projectile experiencing a constant gravitational force and a drag force proportional to its velocity).
- $\eta \in \mathbb{R}$ is the drag coefficient (note that in f_t , the term ηv_t has a negative sign, because it acts in the opposite direction of the velocity).

- $g = (0, -9.8)^T$ is gravity

Dynamics:

- approximate the velocity as constant over the time interval $th \leq \tau \leq (t+1)h$

$$p_{t+1} = p_t + hv_t$$

- approximate force as constant over the same time interval

$$v_{t+1} = v_t + \frac{h}{m}f_t = v_t + \frac{h}{m}(mg - \eta v_t) = (1 - \frac{h\eta}{m})v_t + hg$$

Now we can write this more compactly in matrix-vector form as

$$x_{t+1} = Ax_t + b$$

where

$$A = \begin{bmatrix} 1 & 0 & h & 0 \\ 0 & 1 & 0 & h \\ 0 & 0 & 1 - \frac{h\eta}{m} & 0 \\ 0 & 0 & 0 & 1 - \frac{h\eta}{m} \end{bmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ hg_1 \\ hg_2 \end{pmatrix}$$

Propagating the state through time:

$$\begin{aligned} x_1 &= Ax_0 + b \\ x_2 &= Ax_1 + b = A(Ax_0 + b) + b = A^2x_0 + Ab + b = A^2x_0 + (A + I)b \\ &\vdots \\ x_T &= Ax_{T-1} + b = \dots = A^Tx_0 + (A^{T-1} + \dots + A + I)b \end{aligned}$$

NOTE: In the last equation, in the term A^Tx_0 , the letter T does not represent transpose of the matrix, but the last time point which we represent by T .

Targeting problem

Given:

- initial position p_0
- parameters h, m, η
- flight time : Th
- desired final position (target): p_T

Goal:

- find the initial velocity v_0

We will begin by determining the final state x_T

$$x_T = A^T x_0 + (A^{T-1} + \dots + A + I)b = Fx_0 + j$$

where $F = A^T$ and $j = (A^{T-1} + \dots + A + I)b$.

Note that $F \in \mathbb{R}^{4 \times 4}$ and $j \in \mathbb{R}^4$

The final position can be written as

$$x_T = Fx_0 + j \implies \begin{pmatrix} p_T \\ v_T \end{pmatrix} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} \begin{pmatrix} p_0 \\ v_0 \end{pmatrix} + \begin{pmatrix} j_1 \\ j_2 \end{pmatrix}$$

where $j = (j_1, j_2) \in \mathbb{R}^4$ and $j_1, j_2 \in \mathbb{R}^2$.

We can easily derive that

$$p_T = F_{11}p_0 + F_{12}v_0 + j_1 \implies p_T = Cv_0 + d$$

where $C = F_{12}$ and $d = F_{11}p_0 + j_1$.

So we can find the initial velocity by solving the linear system:

$$Cv_0 = p_T - d$$

Robust ballistics

- Suppose we have uncertainty in the drag coefficient
- Uncertainty can be modeled as K scenarios
- Each scenario k has its own $A^{(k)}, b^{(k)}, C^{(k)}, d^{(k)}$

Robust targeting

Solve the optimization problem:

$$\min_{v_0} \frac{1}{K} \sum_{k=1}^K \|C^{(k)}v_0 + d^{(k)} - p_T\|_2^2$$

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Formal model: Our model for the projectile's motion includes two key forces: gravity and air drag.

- The total force is modeled as $f_t = mg - \eta v_t$. This is a common approximation for a projectile under gravity with a linear drag force, which acts in the opposite direction of the velocity ($-\eta v_t$).
- $\eta \in \mathbb{R}$ is the drag coefficient
- $g = (0, -9.8)^T$ is the acceleration due to gravity

Dynamics: To model the projectile's motion over time, we use a simple numerical integration method (a forward Euler approximation). We assume that the velocity and force are constant over the time interval h .

- The change in position is based on the current velocity:

$$p_{t+1} = p_t + hv_t$$

- The change in velocity is based on the total force, derived from Newton's second law ($f = ma$):

$$v_{t+1} = v_t + \frac{h}{m}f_t = v_t + \frac{h}{m}(mg - \eta v_t) = (1 - \frac{h\eta}{m})v_t + hg$$

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Robust ballistics

- Suppose we have uncertainty in the drag coefficient, η , or other parameters.
- This uncertainty can be modeled by considering K different scenarios, each with its own set of parameters.
- Each scenario k has its own corresponding matrices and vectors: $A^{(k)}, b^{(k)}, C^{(k)}, d^{(k)}$.

Robust targeting

The goal of robust targeting is to find a single initial velocity, v_0 , that performs well across all K scenarios. This is achieved by solving a ****least squares optimization problem**** that minimizes the average squared error.

$$\min_{v_0} \frac{1}{K} \sum_{k=1}^K \|C^{(k)}v_0 + d^{(k)} - p_T\|_2^2$$

This objective function represents the average of the squared distances between the predicted final position ($C^{(k)}v_0 + d^{(k)}$) and the desired target position (p_T), for each scenario. By minimizing this sum, we find a v_0 that is a good compromise across all uncertain scenarios.